

Sprint 4: Executable Quote Engine

How a safe reference rate becomes an immutable customer offer

In progress

Current status: ADR-009 quote economics are approved. The quote policy and immutable quote schema are implemented and verified on the isolated Railway test database. Domain calculation, repository, API, and production migration remain in progress.

Connection to Sprint 3

Sprint 3 stores an accepted reference rate and every provider fact behind it. Sprint 4 will select a fresh snapshot, add the configured spread and purchase fees, calculate exact fiat and crypto amounts, and freeze the result until expiry.

Planned Flow

```
accepted Sprint 3 reference snapshot
      + versioned quote policy
      + total fiat debit
          v
fees -> trade amount -> spread-adjusted rate
          v
immutable executable quote with short expiry
          v
Sendaza displays it; Sprint 6 later reserves inventory
```

Approved Economic Rule

The request supplies total fiat debit. Fixed and percentage fees are deducted before conversion. Percentage fee rounds up to fiat atomic units; destination crypto rounds down to asset atomic units. The source snapshot expiry limits quote expiry. A quote never locks Sendaza money or reserves SLE inventory.

Planned Evidence

- Exact fixed, percentage, combined, and zero-fee tests.
- Limit, precision, rounding, and tiny-amount boundaries.
- Expiry and configuration-change evidence.

- Idempotency and concurrent creation tests.
- Real PostgreSQL atomicity and immutability tests.
- Private API contract and stable error tests.

Current Limitations

The quote tables exist only in `sle_test` ; production migration requires separate approval. No quote calculation service, repository, or endpoint exists yet. Production pricing remains inactive, so no executable quote can currently be issued.

Glossary

Total debit: all fiat the customer authorizes. **Trade amount:** total debit minus fees. **Spread:** configured adjustment from reference rate to customer rate. **Atomic unit:** smallest supported unit of a currency or asset. **Executable quote:** immutable short-lived economics that a later purchase may consume.

Started 1 September 2026. Sprint 4 remains in progress.